

Academic Position

Michigan State University, East Lansing, MI

Postdoctoral Research Associate

Postdoctoral mentor: Prof. Andrew J. Christlieb

Aug. 2024 – present

Education

Missouri University of Science and Technology, Rolla, MO

Ph.D. in Mathematics (Computational and Applied Mathematics Emphasis)

Advisor: Prof. Yanzhi Zhang

Aug. 2019 – May 2024

Shandong University, Jinan, Shandong, China

M.S. in Computational Mathematics

Advisor: Prof. Fuzheng Gao

Sep. 2016 – Jun. 2019

Anhui University of Technology, Ma'anshan, Anhui, China

B.S. in Mathematics and Applied Mathematics

Sep. 2012 – Jul. 2016

Research Interests

- Kinetic equations and structure-preserving numerical methods
- Hyperbolic and well-balanced shallow-water moment models
- Scientific machine learning for moment closures and reduced-order modeling
- Nonlocal and fractional PDEs: modeling, analysis, and fast solvers

Preprints and Manuscripts

8. Globally hyperbolic two-dimensional modified shallow water linearized moment equations and well-balanced finite-volume schemes, **S. Zhou**, J. Huang, and A. J. Christlieb (under review).
7. Hyperbolic machine learning moment closures for the Vlasov–Poisson–BGK equations, A. J. Christlieb, S. Gong, J. Huang, N. A. Krupansky, and **S. Zhou** (in preparation; submission planned before October 15, 2026).
6. A deterministic particle method for the relativistic Landau equation, **S. Zhou**, J. Hu, and A. J. Christlieb (in preparation).
5. An accurate and efficient parallel finite difference scheme for fractional diffusion in heterogeneous media, **S. Zhou**, and Y. Zhang (in preparation; submission planned before September 30, 2026).
4. Data-driven approach for the solutions of time-dependent PDEs using convolutional neural network, **S. Zhou**, Y. Li, and Y. Zhang, to be submitted (2026).
3. Nonlocal modeling of spatial fractional diffusion with truncated interaction domains and truncated kernel function singularity, **S. Zhou**, Y. Zhang, and M. Gunzburger, under review (2025).
2. Moment-enhanced shallow-water equations with an effective wall closure for no-slip bottoms, **S. Zhou**, J. Huang, and A. J. Christlieb, in revision (2025).
1. Analytical and computational aspects of the high-order fractional Laplacian, **S. Zhou**, J. P. Borthagaray, Y. Wu, and Y. Zhang, preprint (2023).

Published Papers

5. Convolutional neural network-based reduced-order modeling for parametric nonlocal PDEs, Y. Wang, **S. Zhou**, and Y. Zhang, *Comput. Methods Appl. Mech. Eng.*, 444 (2025), pp. 118084.
4. Fourier pseudospectral methods for the spatial variable-order fractional wave equations, X. Zhao, Y. Zhang, **S. Zhou**, *Comput. Phys. Commun.*, 318 (2025), pp. 109876.
3. Parametric model reduction with convolutional neural networks, Y. Wang, **S. Zhou**, and Y. Zhang, *Int. J. Numer. Anal. Mod.*, 21 (2024), pp. 716-738.
2. A novel and simple spectral method for nonlocal PDEs with the fractional Laplacian, **S. Zhou** and Y. Zhang, *Comput. Math. Appl.*, 168 (2024), pp.133–147.
1. Weak Galerkin finite element method with second-order accuracy in time for parabolic problems, **S. Zhou**, F. Gao, B. Li, and Z. Sun, *Appl. Math. Lett.*, 90 (2019), pp. 118–123.

Teaching

Recitation Instructor, MTH 235: Differential Equations (MSU) *Fall 2026*
Recitation Instructor, MTH 133: Calculus II (MSU) *Fall 2025*
140 students in total; **Supportive learning atmosphere: 4.13/5; respectful treatment: 4.46/5.**
Instructor, MATH 3304: Elementary Differential Equations (Missouri S&T) *Spring 2023*
98 students in total; **Effectiveness average: 3.13/4.**
Lab Instructor, MATH 1214: Mathematics Calculus for Engineers I (Missouri S&T) *Fall 2020*
61 students in total; **Effectiveness average: 3.5/4.**

Talks and Posters

18. Hyperbolic Machine Learning Moment Closures for the Vlasov–Poisson–BGK Equations, SIAM Conference on Computational Science and Engineering (CSE27), Pittsburgh, PA, 2027/2
17. Hyperbolic Machine Learning Moment Closures for the Vlasov–Poisson–BGK Equations, 2026 SIAM New York–New Jersey–Pennsylvania Section Conference (NNP26), Rutgers University, New Brunswick, NJ, 2026/10
16. A Deterministic Particle Method for the Relativistic Landau Equation, 2026 SIAM Annual Meeting (AN26), Cleveland, OH, 2026/7
15. A deterministic particle method for the relativistic Landau equation, Midwest Numerical Analysis Day 2026, University of Notre Dame, Notre Dame, IN, 2026/4
14. A deterministic particle method for the relativistic Landau equation, 28th International Conference on Numerical Simulation of Plasmas (ICNSP 2025), UCI, Irvine, CA, 2025/11
13. Moment-enhanced shallow water equations for non-slip boundary conditions, 2025 SIAM New York–New Jersey–Pennsylvania Section Conference, Penn State, State College, PA, 2025/10
12. Data-driven modeling of time-dependent nonlocal PDEs, CHaRMNET 2024 Annual Meeting, Virginia Tech, Blacksburg, VA, 2024/12
11. A novel and simple spectral method for nonlocal PDEs with fractional Laplacian, The 8th Annual Meeting of SIAM Central States Section, Lincoln, NE, 2023/10
10. A novel and simple spectral method for nonlocal PDEs with fractional Laplacian, Graduate Seminar at Missouri University of Science and Technology, Rolla, MO, 2023/4
9. Numerical studies on the high-order fractional Laplacian, Pi Day at Missouri University of Science and Technology, Rolla, MO, 2023/3

8. Accurate and efficient spectral method for fractional wave equations, Numerical Analysis Seminar at University of Pittsburgh, Pittsburgh, PA, 2023/2
7. Accurate and efficient spectral method for fractional wave equations, The 7th Annual Meeting of SIAM Central States Section, Stillwater, OK, 2022/10
6. Numerical studies on the high-order fractional Laplacian, Theoretical and Applied Aspects for nonlocal Models - Workshop of Banff International Research Station (BIRS), Online, 2022/7
5. Numerical methods for acoustic wave equations, Graduate Seminar at Missouri University of Science and Technology, Rolla, MO, 2022/4
4. Numerical studies on the high-order fractional Laplacian, The University of Kansas Numerical Analysis Day 2022, Lawrence, KS, 2022/3
3. Numerical studies on the high-order fractional Laplacian, Continuum Mechanics Seminar at University of Nebraska-Lincoln, Online, 2022/3
2. Numerical studies on the high-order fractional Laplacian, 4th Annual Meeting of the SIAM Texas-Louisiana Section, South Padre Island, TX, 2021/11
1. Numerical studies on the high-order fractional Laplacian, Midwest Numerical Analysis Day 2021, Rolla, MO, 2021/10

Referee for Professional Journals

- Zeitschrift für angewandte Mathematik und Physik (ZAMP)
- Mathematics and Computers in Simulation

Skills

- Numerical analysis and modeling: nonlocal and fractional PDEs, hyperbolic moment systems, kinetic equations, and reduced-order models.
- Numerical methods: Fourier spectral and pseudospectral methods; finite-difference, finite-volume, and Galerkin methods; well-balanced and structure-preserving discretizations; deterministic particle methods.
- Scientific machine learning: convolutional neural networks for reduced-order modeling and time-dependent PDE solvers; hyperbolicity-aware learned moment closures; PyTorch.
- Research computing: C++, MATLAB, Python, and Fortran; FFT-based and matrix-free solvers; OpenMP and Slurm workflows; reproducible numerical verification; $\text{\LaTeX}$ and Git.

Professional Membership

- Society for Industrial and Applied Mathematics (SIAM) *Jan. 2021 – present*

Awards

- Gaoxiong Gan Scholarship, Missouri S&T, 2022, 2023, & 2024
- Paul W. Elie Graduate Research Award, Missouri S&T, 2022 & 2023
- 8th Annual Meeting of SIAM Central State Section Travel Award, 2023
- First place at the Pi Day Celebration Poster Session, Missouri S&T, 2023
- Graduate Education Travel Fund, Missouri S&T, 2023
- Nonlocal School on Fractional Equations (NSFE) Travel Award, 2022
- Mathematics and Statistics Alumni Endowed Scholarship, Missouri S&T, 2022
- 7th Annual Meeting of SIAM Central State Section Travel Award, 2022